CTS — Aptitude Test 2

Questions 26–50 of 100 · 25 questions

Question 26. The difference of 2 numbers is 8 and the difference of their squares is 160. Find the numbers?

Solution:
Segregate the binary numbers in groups
of three from the right end:
(11 111 011)2
Octal equivalent of 011 = 3
Octal equivalent of 111 = 7
Octal equivalent of 11 = 3
(11111011)2 = (373)8

Solution:
Let the two numbers be X and X – 8
X2 – (X – 8)2 = 160
X2 – (X2 + 64 – 16X) = 160
X2 – X2 – 64 + 16X = 160
16X = 224 ; X = 14
The two numbers are 14 and 6

Question 27. The total number of numbers that are divisible by either 2 or 3 between 100 and 200(inclusive) are:

Solution:

Solution:
The number of games I won = 9/3 = 3
Number of games my friends won
= (9/3) + (12/3) = 7
Total games played = 10

Question 28. Convert Binary to Octal (11111011)2= ( )8

Question 29. I and two of my friends were playing a game. For each win I get Rs 3. totally I had three wins. Player 2 got Rs9 and player 3 got Rs 12. How many games had been played.

Question 30. Three wheels make 36, 24, 60 revolution/min. Each has a black mark on it. It is aligned at the start. When does it align again for the first time?

Solution:
Time taken for first wheel to complete
one revolution = 60/36 minutes
Time

taken
for
second
wheel
to
complete one revolution = 60/24 mins
Time taken for third wheel to complete
one revolution = 60/60 = 1 minute.
The time taken for all the black marks
to align is
= LCM[(60/36), (60/24), 1]
= LCM(5/3 , 5/2, 1)
= 5 minutes

Question 31. Number of prime factors of 307 x 225 x 3411 ?

Solution:
307 = (2 x 3 x 5)7 = 27x37x57

225 = (2 x 11)5 = 25 x 115

3411 = (2 x 17)11 = 211 x 1711

307 x 225 x 3411 = 223x 37x 57x 115x 1711

The number of prime factors = 5

Question 32. What is the remainder wen 2^35 is divided by 5?

Solution:
To find the reminder when divided by 5,
we have to know only the unit digit of
the numerator.
Unit digit of 235 = 8
So the reminder of 235/5 = 3

Question 33. What is the remainder wen 825 is divided by 7?

Solution:
Reminder of [xn/(x-1)] = 1
So reminder of 825/7
= 1

Question 34. 8^2x / 8^5 = 8^7. Find x

Solution:
82x / 85 = 8(2x – 5)

8(2x – 5) = 87

2x – 5 = 7
2x = 12
x = 6

Question 35. 2^81 when divided by 6, what is the remainder?

Solution:
Reminder of 21/6 = 2
Reminder of 22/6 = 4
Reminder of 23/6 = 2
Reminder of 24/6 = 4
So, (2^odd/6) = reminder = 2
(2^even/6) = reminder = 4
Therefore, the reminder of
281/6 = 2

Question 36. Five farmers have 7,9,11,13 & 14 apple trees respectively in their orchards. Last year each of them discovered that every tree in their own orchard bore exactly the same no of apples. Further if the 3rd farmer gives one apple to the 1st & the 5th gives 3 to each of the 2nd & the 4th they would all exactly have the same no of apples. What were the yields per tree in the orchards of the 3rd & 4th farmers?

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Solution:
Assume that the number of apples bore in each farmers trees = A, B, C, D, E.
After exchanging all the apples, the number of apples left with each farmer
= A + 1, B+3, C – 1, D + 3, E – 6
Now the number of apples with all farmers are same.
Therefore,
C – 1 = D + 3
We know that C had 11 trees and D had 13 trees. So, C is a multiple of 11 and D is a multiple of 13.
11x – 1 = 13y + 3
By trial and error method, we can find that x = 11 and y = 9

Question 37. Four persons can cross a bridge in 3,7,13,17 minutes. Only two can cross at a time. Find the minimum time taken by the four to cross the bridge?

Case 1:
Weighing number 1:
Take the weight balance and keep any two groups on each plate.
If one of the plate goes down, take that plate for the next weighing.
Weighing number 2:
Out of the three balls chosen from first weighing, take any two and keep one on each plate. The plate which goes down carries the defected ball. If the plates did not go down, the ball which was kept outside was the defected one.
Case 2:
Weighing number 1:
Take the weight balance and keep any two groups on each plate.
If none of the plate goes down, take the group of balls which was kept outside for the next weighing.
Weighing number 2:
Out of the three balls chosen from first weighing, take any two and keep one on each plate. The plate which goes down carries the defected ball. If the plates did not go down, the ball which was kept outside was the defected one.
Minimum number of measurements = 2

Solution:
Let 17 minute person and 13 minute person cross the bridge initially.
In 13 minutes, one person will cross and the 17 minute person will be on the bridge.
At the end of 13 minutes, let 7 minute person start walking on the bridge so that there are only two persons on the bridge now.
When 17 minutes is completed, the two persons would have crossed the bridge and the 7 minutes person is still on the bridge with 3 minutes remaining to cross.
At this time the person with 3 minutes may start crossing the bridge.
At the end of 20 minutes, all the persons would have crossed the bridge.
Minimum time taken = 20 minutes.

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Question 38. I had Rs100 and I play. If I win I will have Rs110 and if I lose I will have Rs90. at the end I have 2 wins and 2 loses. How much do I have?

Question 39. There are 9 balls which of one are defective. Find the minimum chance to take that defective one.

Solution:
The 9 balls are separated as groups of three.
Now we have 3 groups of 3 balls each.

and the 1/7th bar and 2/7th bar will be bought back.
On the 5th day, 1/7th bar will be given.
On 6th day, 2/7th bar will be given and the 1/7th bar will be bought back.
On 7th day, 1/7th bar will be given back.

Question 40. A worker is to perform work for you for seven straight days. In return for his work, you will pay him 1/7th of a bar of gold per day. The worker requires a daily payment of 1/7th of the bar of gold. What and where are the fewest number of cuts to the bar of gold that will allow you to pay him 1/7th each day?

Solution:
The prize percentage is 10%.
Amount after first win
= 100 + 10%(100) = 110
Amount after second win
= 110 + 10%(110) = 121
Amount after first lose
= 121 – 10%(121) = 108.9
Amount after second lose
= 108.9 – 10%(108.9) = 98.01

Solution:
The minimum number of cuts = 2
Explanation:
On day one 1/7th of the bar is cut and given to the worker.
On second day, 2/7th of the bar is cut and given to the worker and you must get back the 1/7th bar given on the first day.
Third day, the 1/7th bar will be given again.
Fourth day, 4/7th of the bar which was remaining will be given to the worker

Question 41. It was 4 ‘o’ clock in the evening. Shilu was staring at the new watch that was presented by her Dad two day's ago. She was trying to measure the exact time between 4 and 5 ‘o’ clock during which the hands of the watch point in opposite directions forming a straight line. What

would be that time?

Solution:
The two hands will be 30 minutes apart from each other when they are pointing in exact opposite directions.
The formula to find out the time is:
T = [(12/11)(5h + 30)]
Where h = 4.
T = [(12/11)(20+30)]
T = 54.54 minutes past 4 ‘o’ clock
T = 4 hours 54 minutes 33 seconds

Question 42. In a clock (Conventional clock with numbers 1 to 12 in order)is cut into 3 pieces such that the sum of each piece is in arithmetic progression with a common difference of 1. What is the count of numbers in each piece? and what are the numbers?

Solution:
Sum of all the numbers from 1 to 12 is
12(12+1)/2 = 78.
So sum of the three groups after splitting will be 78.
Since the sum of each group is in arithmetic progression, one of the group has sum as the average of three groups.
Average = 78/3 = 26.
Sum of the groups are: 25, 26 and 27
The hour groups are
25 = 3 + 4 + 5 +6 + 7
26 = 11 + 12 + 1 + 2
27 = 8 + 8 + 10
We have to find out using trial method.

Question 43. Pooja told Narmadha, "I am four times as old as you were when I was your present age and also I am 9 years older than you" . What is Pooja's age?

Solution:
Let Narmadha’s present age be N.
Let Pooja’s present age be P.
N = P – 9
---(1)
P – 9 = 4(N – 9)
---(2)
Sub (1) in (2)
P – 9 = 4[(P – 9) – 9]

P – 9 = 4P – 72
3P = 63
P = 21 ■Age of Pooja = 21

Question 44. If I am twice as old as he was when I was as old as him, sum of our ages is 42. Find my present age?

Solution:
Let my present age be X.
Let his present age be Y.
His age when I was at his age
= Y – (X – Y)
X = 2(Y – (X – Y))
X = 4Y – 2X
3X = 4Y and
X + Y = 42
4X + 4Y = 168
4X + 3X = 168 – 7X = 168
X = 24.

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Question 45. When I was married 10 years back my wife was the sixth member of my family. Now I have a baby. Today my father was dead and I had a new baby. now the average age of my family is the same as that when I was married. Find the age of my father when I was married.

Solution:
Let the average age of my family when I got married be X.
Total age of the family 10 years ago
= 6X.
Total age of the family now with the new born baby = 6X + 60 + 0 = 6X + 60
Average age of the family with my father died = X

Question 46. A boy asks his father, "What is the age of grandfather?" Then father replied "He will be X years old in the year X2". The father was talking about the 21st century. In which year was the grandfather born?

Solution:
The only perfect square in between 2000 and 2100 is 2025.
2025 is the square of 45.
So in the year 2025, the grandfather will be 45 years old.
The grandfather was born in the year 1980.

Question 47. A trend was observed in the growth of population in Saya Islands. The population tripled every month. Initially the population of Saya Islands was 100. What would be population after 4 months ?

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Solution:
Let the Share of B = X
Ratio between share of A and B is (3/5)X : X = 3 : 5
Share is directly proportional to investment and duration of investment.
A invested for 9 months and B invested for 12 months
3 : 5 = 27000 x 9 : B x 12
12B = 27000 x 9 x 5 / 3
B = 33750

Solution:
Population after one month = 300
Population after two months = 900
Population after three months = 2700
Population after four months = 8100

Question 48. The speeds of two trains are in the ratio 5:3.If the speed of the first train is 350km in 2hours. Then the speed of second train is?

Question 49. In a class the ratio of boys and girls is 5:6. If 25% of boys and 20% of girls are scholarship holders, find the percentage of students who are not scholarship holders?

Let the father's age be F.
After the death of father the total age of family = 6X + 60 – F
So, [6X + 60 - F]/6 = X
F = 60.
Age of father when he died = 60

Solution:
Let us assume that there are 100 boys and 120 girls. (It is 5 : 6 ratio).
25 boys are scholarship holders and 75are not.
24 girls are scholarship holders and 96 are not.
Total number of students not having scholarship = 75 + 96 = 171.
Total number of students = 220
Percentage of students with no scholarship
= (171/220)100
= 77.72%

Question 50. A joined in a partnership with B after 3 months by investing Rs. 27000. The profit of A is 3/5th of B's share at the end of one year. What was the amount invested by B?